Property-IX

(i) $\sin^{-1} x = \cos^{-1}\sqrt{1-x^2}$, if $0 \le x \le 1$. (ii) $\sin^{-1} x = -\cos^{-1}\sqrt{1-x^2}$, if $-1 \le x < 0$.

(ii) $\sin^{-1} x = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$, if $-1 < x < 1$. (iv) $\cos^{-1} x = \sin^{-1}\sqrt{1-x^2}$, if $0 \le x \le 1$.

(v) $\cos^{-1} x = \pi - \sin^{-1}\sqrt{1-x^2}$, if $-1 \le x < 0$. (vi) $\tan^{-1} x = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$, if $x > 0$.

Property-X

(i) $3\sin^{-1} x = \sin^{-1}(3x - 4x^3)$, $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$. (ii) $3\cos^{-1} x = \cos^{-1}(4x^3 - 3x)$, $x \in \left[\frac{1}{2}, 1\right]$.

ICT CORNER

https://ggbm.at/vchq92pg or Scan the QR Code

Open the Browser, type the URL Link given below (or) Scan the QR code. GeoGebra work book named "12th Standard Mathematics" will open. In the left side of the work book there are many chapters related to your text book. Click on the chapter named "Inverse Trigonometric Functions". You can see several work sheets related to the chapter. Select the work sheet "Graph of Inverse Trigonometric Functions"

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